

Steam reforming of hydrocarbons. A historical perspective

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ABSTRACT

After the industrial break-through of steam reforming, main problems were linked to sulphur poisoning, carbon formation and attempts to minimize the reformer size. The understanding of the catalysis followed the industrial developments with most studies carried out by industrial groups. Today, the hydrogen economy has created great interest in academia and industry to explore the limits of the reaction.

1. INDUSTRIAL DEVELOPMENTS

The steam reforming reaction was introduced to industry in 1930 more than 70 years ago [1, 2, 3, 4], although the reaction had not been subject to many scientific studies [1, 3]. The reforming process was adopted mainly in the US where natural gas was easily available as feedstock whereas reformers in Europe were initially reduced to operate on propane and LPG.

The industrial break-through of the steam reforming technology occurred in 1962 when ICI succeeded in starting two tubular reformers operating at pressure (15 bar) and using naphtha as feedstock [5]. The first reformer designed by Topsoe was started in 1956 and the first naphtha reformer in 1965 followed by a hydrogen plant in 1966 operating at 42 bar [1, 4]. The introduction of high pressure reforming meant that the energy consumption of the ammonia synthesis could be decreased significantly [6] which was further decreased to less than 28 GJ/t by integration of the steam turbine cycle and process and application of better catalysts. As a result, a number of natural gas based ammonia plants were built in the US primarily by MW Kellogg in the mid sixties [7].

At the same time in Europe, steam reforming of naphtha became the key technology in the build-up of the UK towns gas industry replacing low pressure gasification processes. Apart from the tubular reforming technology provided by ICI [5] and Topsøe [8] low temperature steam reforming of naphtha in an adiabatic reactor was used to produce methane-rich gas [1, 3]. This process was

pioneered by British Gas [9] originally using methanol as fuel. The steam reforming of naphtha was used also for ammonia plants in countries like India [10] and for hydrogen in refineries in particular in Japan [11], where natural gas was not available. The steam reforming process also became the technology for syngas for methanol and other petrochemical products, oxo-alcohols, acetic acid, etc. [1, 12]. During the energy crisis in the 1970'ties, low temperature steam reforming followed by methanation was used to convert naphtha into substitute natural gas [4]. During the 1970'ties and 80'ties, it was predicted that the steam reforming of natural gas would form the basis for a new industry for direct reduction of iron-ore [13, 14] which however did not fully develop. Steam reforming was also applied to supply hydrogen for a series of prototype fuel cell plants started in the same period [15].

Today, the steam reforming process is well-established for the production of hydrogen and syngas for large scale commodity chemicals [16]. The need for hydrogen is increasing in refineries due to demands for low sulphur, low aromatics and hydrocracking of heavy feedstocks [17]. The steam reforming of natural gas and liquid fuels (logistic fuels) may play a key role for the introduction of a hydrogen economy. Hydrogen for fuel cells will be an important element [18], but steam reforming is also a key reaction in the anode chamber of high temperature fuel cell [19]. In spite of efforts to produce hydrogen by schemes involving solar energy, bio-fuels etc., steam reforming of natural gas remains the cheapest route to hydrogen in the near term [17].

The steam reforming reaction may be applied as a mean for chemical recuperation [20] as pioneered at the ADAM/EVA system [20] for nuclear energy and now considered for solar energy. Chemical recuperation may also be applied with high temperature fuel cells (SOFC) coupled to gas turbine schemes [16].

The manufacture of syngas in the first step in the conversion of remote natural gas into liquids for instance via the FT synthesis. However, for large scale syngas plants, oxygen-blown autothermal reforming becomes more feasible [16]. Also for very small scale operation, oxidative processes might become more economic because of simplicity and compactness [18].

2. TOWARDS THE LIMITS

Steam reforming involves the risk of *carbon* formation in particular at operation with low steam to carbon ratios or at conditions aiming at low H_2/CO ratios. The potential for carbon formation could be assessed by the principle of equilibrated gas [12]. The mechanism of carbon formation via whisker carbon was studied by primarily electron microscopy [1, 21] and TGA [22, 23].

The thermodynamic limits were surpassed by using noble metals [22, 24] or by using a partly sulphur passivated nickel catalyst [25] as applied in the

SPARG process [26]. Both solutions were introduced to industry during the 1980'ties at conditions at which thermodynamic would predict carbon formation.

When steam reforming of naphtha was introduced in the mid 60'ties, *sulphur* poisoning was a problem [27,28]. The typical event was that the inlet layer was sulphur poisoned which resulted in non-converted naphtha passing to the hot part of the reforming tube where carbon would be formed by pyrolysis and the development of hot tubes [12].

Around 1970, precise knowledge was obtained on the chemisorption of sulphur on nickel – not the least through LEED-studies [29]. It was possible to estimate [12] that the true sulphur limits ca. 10 ppb and not 0.2 ppm (the analytic limit in the 60'ties). For normal naphthas, the sulphur limit could easily be achieved over new catalysts and absorption masses [27].

At first, alkali *promoted catalysts* were used for the tubular steam reforming of naphtha [5, 8], but it was possible to eliminate the potassium promotion and the problems associated with alkali by using active magnesia [1, 30]. At the same time, it was possible to achieve much higher activity because alkali was shown to drastically reduce the activity of the nickel surface [1]. It meant that the alkali-free catalyst could convert heavy naphtha with high contents of aromatics and hence less reactivity [10]. It was shown that if the feedstock was properly desulphurised, the catalyst could also steam reform kerosene and gas oil with the formation of only C₁-components [1, 10, 31]. The sulphur problem was solved with the introduction of a *prereformer* operating as a low temperature adiabatic reformer [16, 31]. The prereformer converts the higher hydrocarbons before entering the preheater to the tubular reformer. Without the higher hydrocarbons, it became possible to preheat to high temperatures without the risk of pyrolysis. This principle was introduced also to natural gas based reformers during the 1980'ties [16, 31]. The preformer was also applied in schemes for high temperature fuel cells [18].

The steam reformer is highly efficient, but the tubular reformer is very costful. Therefore there has been a strong effort to *minimize the size* of the tubular reformer. The steam reforming process may appear straightforward because the product gas is determined by thermodynamic equilibrium. It is however a complex coupling of heat transfer and catalysis and computer models are required to estimate tube wall temperatures and the risk for carbon formation. These models are not too sensitive to reactions kinetics as the gas is close to equilibrium in most of the tube [12].

The first kinetic studies were strongly influenced by diffusion restrictions [2, 12] and it was not until the Temkin group in the 1960'ties [32] studied the reaction on nickel foil that more precise information was achieved. Still empirical kinetics [33] were sufficient to develop highly sophisticated reactor models for the tubular reformer [34, 35].

With proper desulphurisation and the use of a prereformer, the catalyst activity is rarely a limiting factor [16]. The availability of better materials for reformer tubes and a better understanding of the stress problems allowed an increase of the heat transfer and hence a reduction of the number of tubes for a given service. With a prereformer, it is possible to increase the inlet temperature to the tubular reformer and thus reduce the size of reformer. Today reformers are designed for operation at average heat fluxes [17,36] almost two times higher than what was the industrial practice 20 years ago.

It is possible to increase the amount of heat transferred to the process gas in the reformer from about 50% to about 80% of the supplied heat when using a convective heat exchange reformer [16,36]. This resulted in a more compact piece of equipment, and it is possible to design for no-steam export. This type of *convective reformers* were used for fuel cell demonstration plants during the 1970's and 80's [37]. Today, it is applied for minor hydrogen plants [17, 36]. More compact designs can be developed for small scale reformers. It is a challenge to take advantage of the huge surplus of activity by using catalysed hardware and non-tubular designs [38] as explored in micro-channel reformers [17, 39, 40]. Another route may be membrane reforming at low temperature [17, 41].

Steam reforming of hydrocarbons in compact units may become the vehicle for introduction of a *hydrogen economy* as the preferred choice for decentralized units at gas stations or for fuel cells.

3. TOWARDS COLLABORATIVE EFFORTS

The development of the steam reforming process has not been based on initial understanding of the catalysis. The progress was driven mainly by an "inductive approach" with feedback from industrial operation and pilot tests [4, 35]. The initial vague ideas on the mechanism [1, 42, 43] were formulated by industrial groups around 1970 whereas only few academic scientists were involved in studies of the reaction [32, 44, 45, 46]. Research on nickel catalysts was more related to hydrogenolysis [47, 48] and later methanation [49, 50] became of interest in SNG from coal in the late 1970's. In the 1990's, the discussion about the greenhouse effect caused a series of studies on CO₂-reforming [24, 50] although CO₂-reforming may hardly solve the greenhouse problem and that stoichiometric reforming is rarely feasible [16]. Lately, the prospects of a hydrogen economy have caused a number of projects on compact small scale steam reformers [18, 39, 40].

Early steam reforming studies [1] at well-defined conditions aimed at describing parameters influencing the nickel surface area and the specific activity. This was later supplemented by ultra-high vacuum studies of methane activation on well-defined surfaces [51, 52, 53]. Recent studies [16, 54] have led

to a more consistent understanding of the mechanism at the atomic level. DFT-calculations, adsorption studies [54] and in-situ high resolution electron microscopy [55] have shown that step sites play a key role in methane activation as well as for nucleation of carbon. This had led to a more rational approach to catalyst promotion [16]. It has revived old ideas of B5 sites [1, 56]. The development since the late 1980'ties has been characterised by significant contributions from academia [33, 50, 57, 58, 59, 60, 61, 62] and a balanced collaboration between academic and industrial research teams [16, 53, 54, 63, 64, 65, 66, 67].

It was the fundamental understanding of carbon formation and sulphur poisoning in terms of ensemble control [25] which formed the basis for the introduction of the SPARG process in the 1980'ties [26]. The present knowledge at the atomic scale [16, 67] may lead to more advanced reforming systems and it has contributed to the general knowledge of metal catalysis.

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